Effective Rotate: Learning Rotation-Robust Prototype for Aerial Object Detection

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***Abstract*— Aerial images often depict objects with arbitrary orientations, which pose challenges for conventional object detectors to detect and classify. To address this issue, rotation-equivariant convolutional neural networks (CNNs) have been proposed to extract rotation-equivariant features. However, the orientation encoding in these networks is often unstable and noisy, deteriorating detection performance. In this article, we first analyze the rotation-equivariant network. Then, we propose a rotation-robust prototype generation (RPG) method, which consists of two parts, a stabilization module and an enhancement module. In the stabilization module, we generate rotation-robust prototypes to increase the stability of cyclic shifts. In the enhancement module, we use the obtained prototype to improve the response of the features to object semantics. The RPG method can be used as a plug-and-play module in both one-stage and two-stage detectors. With only 30 lines of code, we achieve an average 1% improvement on four challenging datasets, including DOTA-V1.5, DOTA-v1.0, DIOR-R, and HRSC2016.**

***Index Terms*— Aerial images, enhancement module, rotation-equivariant, stabilization module.**

I. Introduction

AERIAL object detection is an important subfield of object detection in computer vision, which focuses on detecting and classifying objects in aerial images, especially satellite and drone images. It has wide-ranging applications in various fields, including crop and environmental monitoring, military reconnaissance, resource exploration, and more [35]. Compared to object detection in natural images, aerial object detection presents a unique challenge in that objects in aerial images can have arbitrary orientations, making it difficult to achieve optimal results using generic object detectors. To address this issue, some methods [30], [31], [32] use oriented bounding boxes (OBBs) to locate and classify aerial objects. In two-stage networks, existing methods [9], [49] transform the horizontal region of interest (RoI) to the rotational RoI. Despite their progress, conventional convolutional neural networks (CNNs) face limitations in explicitly modeling orientation variations, requiring a large amount of rotated data to train a reliable object detector. Recently, in ReDet [17], the authors proposed rotation-equivariant networks to try to handle this problem.

However, the orientation encoding using rotation-equivariant features is typically unstable. In addition, in some cases, certain channels may not activate object features, as illustrated in Fig. 1. This leads to when the detector directly uses it as a backbone, the exploitation of the orientation encoding is limited.

In this article, we first analyze the limitation of the rotation-equivariant feature encoding paradigm, whose main idea is to use cyclical shifts on multiple independent channel groups, i.e., the position of each channel within the channel group is shifted, as shown in Fig. 2 (left). However, the cyclic shift could be unstable, resulting in invalid orientation encoding. Based on the above observation, we propose the rotation-robust prototype generation (RPG) method to obtain rotation-robust features and use these to augment the original features to activate the object response on the channel. The proposed RPG contains two main parts, a stabilization module and an enhancement module. In the stabilization module, we first slice the rotation-equivariant features in the channel dimension and later aggregate them to obtain the rotation-robust prototype, which is able to suppress the effect of outliers in the original features and contains a more stable cyclic shift. In the enhancement module, prototype features are used to augment the original features. Each feature sliced in the stabilization module is first connected to the prototype to obtain valid semantic information and then fused with itself. By obtaining a more robust rotation encoding from the prototype, the orientation representation of the features can be enhanced. At the same time, since the prototype contains all the semantic information of the sliced features and enables the features to learn the correlation between channels to suppress their own outliers.

We evaluate the proposed method through extensive experiments on several challenging aerial image datasets, including DOTA-v1.0, DIOR-R, and HRSC2016. Our method outperforms state-of-the-art methods under both one-stage and two-stage settings. In summary, our contributions are as follows.

1) We propose an effective method called RPG for the aerial object detection task, which could provide more accurate orientation encoding while being robust to outliers.

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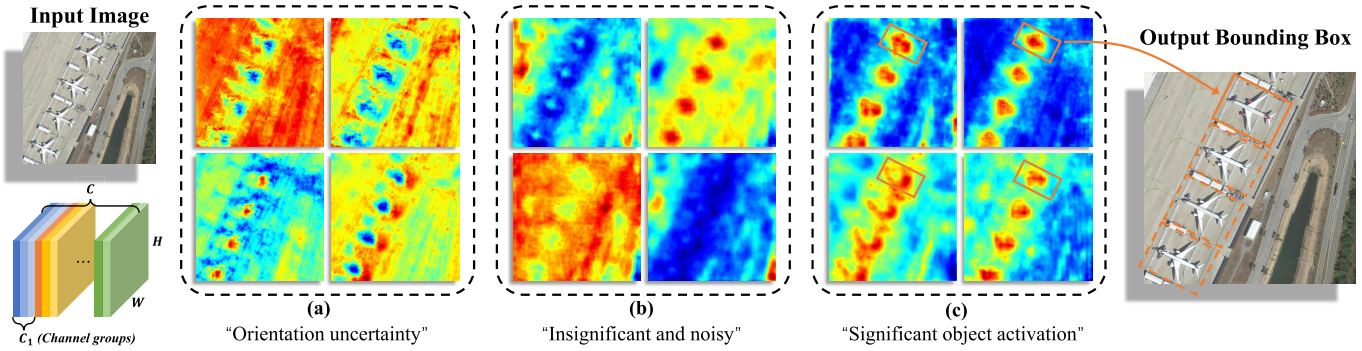


Fig. 1. We compare the feature maps in (a) conventional CNNs, (b) equivariant CNNs, and (c) our proposed RPG model. Compared to conventional and equivariant CNNs, the proposed RPG model could better encode the orientation semantics. Meanwhile, our RPG model addresses the partial activation problem of equivalent CNNs, behaving better in complex scenes.

2) The RPG method could be used as an easy plug-in module for both one- and two-stage, anchor-based and anchor-free detectors with extremely simple implementation.
3) The RPG method achieves 1% improvement on various datasets including HRSC2016, DIOR-R, DOTA-v1.5 and DOTA-v1.0 datasets with only 30 lines of code.

The rest of the article is organized as follows. Section II introduces the related works. Section III introduces the channel cyclic shift. Section IV introduces the detail of our RPG model. In Section V, the experimental results and analysis are reported on challenging DOTA, DIOR-R, and HRSC2016 datasets. Finally, the conclusion is made in Section VI.

II. Related Works

With the rapid advancement of machine learning, particularly deep learning, significant strides have been made in object detection in recent years. Object detection methods can be broadly categorized into two groups: two-stage detectors and one-stage detectors. Two-stage detectors [14], [15], [18], [41] typically involve generating a sparse set of RoI proposals in the first stage, followed by RoI-wise bounding box regression and object classification in the second stage. On the other hand, one-stage detectors such as YOLO [40] and SSD [30] directly detect objects without the need for an explicit RoI generation step. However, one-stage detectors often face challenges due to the imbalanced foreground-background class distribution, resulting in inferior performance compared to two-stage detectors. To mitigate this issue, techniques like focal loss [29] can be employed. In addition, anchor-free detectors [24], [57], [59] approach object detection as a point detection problem, eliminating the need for anchor-based computations and typically offering faster inference speed.

A. Oriented Object Detection

With the development of deep learning, general object detection has achieved great success. However, when considering aerial object detection, objects are captured in a vertical position. Therefore, the general object detectors [2], [30], [40], [41], [42] use the horizontal bounding boxes (HBBs) to locate the objects. Unfortunately, it causes serious dislocation in aerial images, that is, a box will contain multiple objects. Oriented object detectors locate and classify aerial objects with OBBs to address this problem. In the early methods [31], [32], [34], the preset rotation anchors with different aspect ratios, sizes, and angles are used, but the abundant anchors cause massive computation and memory footprint. Ding et al. [9] proposed the RoI transformer to convert the horizontal RoI into rotating RoI after RPN. Although the RoI transformer reduces the number of anchors, it also brings about costly computation. Xie et al. [49] used the external horizontal boxes to predict the offset, which makes the network achieve good performance without additional computation and redundancy. There is also a way to encode the information of the rotating anchor boxes into the features. S^2ANet [16] used alignment convolution to encode the predicted rotation anchors into the feature maps for the next stage, which realizes the alignment of features in different network stages. Li et al. [27] used the adaptive point method to detect objects, and proposed three directional conversion functions to facilitate accurate classification and localization. Xu et al. [50] proposed a dynamic prior and a coarse-to-fine assigner method to address the mismatch and imbalance issues induced by limited features of extreme targets. Doloriel and Cajote et al. [11] accurately detected small oriented objects by enhancing the classification and regression tasks of the oriented object detection model. RFLA [51] proposed a label assignment strategy based on Gaussian receptive fields (RFLA), aiming to better identify tiny objects by measuring the similarity between receptive fields and actual objects. Hou et al. [21] proposed flexible shape-adaptive selection (SA-S) and shape-adaptive measurement (SA-M) strategies to address the oversight of object shape information and the inability to simultaneously apply to anchor-free or anchor-based architectures in existing methods. GGHL [22] utilizes an anchor-free object-adaptation label assignment strategy and an OBB representation component to adaptively adjust Gaussian center priors' weights through neural network learning. In addition, it improves via joint optimization loss to optimize the best results for various subtasks. Some methods [33], [54], [56] enhanced the network performance by improving the loss function.

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The methods discussed above modified the network architecture to improve its performance in detecting oriented objects. However, conventional CNNs do not explicitly encode object orientation, which limits the detector's ability to interpret orientation. To address this issue, some methods have focused on extracting orientation features.

B. Feature Extraction Methods

Methods applying rotation invariance and equivariance characters for rotational object detection have shown great progress recently. Cheng et al. [6] introduced rotation-invariant layers and incorporated a rotational regularization term to design a rotation-invariant CNN. Feng et al. [12] was the first to construct a rotation-invariant aerial object detection network under weakly supervised learning, optimizing instance refinement and rotation invariance in an end-to-end manner. However, both of these methods utilized data augmentation to learn rotation-invariant features, leading to weak generalization. On the other hand, ReDet [17] leveraged a rotation-equivariant network (E2CNN) to acquire rotation-equivariant features, then used channel alignment modules and rotational ROIs to obtain rotation-invariant features. While ReDet generates rotation-invariant features in its network architecture, it does not address the instability issue of rotation-equivariant features. This article addresses this problem by introducing an effective RPG module.

The previous methods encode the orientation of an object, but it is often unstable. In the designed RPG module, we have developed a method to obtain more stable object encoding, for more effective learning.

C. Rotation-Equivariant Networks

Cohen and Welling [7] first proposed the group convolution network to achieve rotation equivariance, and extended the traditional convolution to a larger discrete group, but the discrete group cannot achieve a certain orientation of equivariance. Worrall et al. [47] introduced harmonic filters to produce equivariant features in the continuous domain, but it is difficult to build a deep network, thus the ability of feature extraction is limited. Cohen and Welling [8] introduced the concept of the fiber bundle and redesigned the equivariant model by using the group representation theory. Weiler and Cesa [45] proposed a general group equivariant network framework for planar Euclidean space, which makes the application of rotation equivariance more convenient.

III. Rethinking Channel Cyclic Shift

Section III-A first introduces rotation-equivariant convolutions and rotation-equivariant networks. In Section III-B, the cyclic shift phenomenon of rotation-equivariant features is explained. We model it using a cyclic matrix in Section III-C and highlight the consequences of its instability.

A. Preliminaries

1) Rotation-Equivariant Convolution: Recent methods [7], [20], [46] extend the CNN operations to large group G, achieving both translation and rotation-equivariance in networks. The group G can be defined as the semidirect product of the translation group $(\mathbb{R}^2, +)$ and the rotation group. We formulate it as $G \cong (\mathbb{R}^2, +) \rtimes H$, where $(\mathbb{R}^2, +)$ and H denote the translation group and rotation group, respectively. Let T_g denote an action of the rotation group and M denote feature maps. The rotation-equivariant convolution can be defined as

$$[[T_g M] * \psi](g) = [T_g[M * \psi]](g), \quad g \in G \tag{1}$$

where $\psi : \mathbb{Z}^2 \to \mathbb{R}^2$ denotes the mapping of the convolution filter and $*$ is the convolution operation.

2) Rotation-Equivariance in CNNs: Equivariance in the network refers to feeding a changed image to networks is the same as changing feature maps of the original image. Different from conventional CNNs, rotation-equivariant networks share both translation and rotation weights. Let Φ denote the rotation-equivariant network under the group G and X denote the input image. If we apply the rotation transformation T_r to the input X and feed it to the network Φ, T_r can be preserved by the network, as

$$\Phi(T_r(X)) = T_r \Phi(X), \Phi(X) \in G \tag{2}$$

where T_r on the left operates in the image domain, while the T_r on the right operates in the feature domain. Even though they both represent rotational changes, their specific meanings differ due to their distinct scopes of operation [17].

B. Phenomena

The features in the rotation-equivariant CNNs are named rotation-equivariant features. It has the cyclic shift phenomena [8], i.e., the feature cyclically shifts within each channel group when the input image X rotates, shown in Fig. 2. Therefore, in (2), when T_r operates in the feature domain, it represents a rotation transformation in spatial space and a shift transformation in channel space. Let P_r^n denote a spatial rotation transformation with the angle of $n \times (2\pi/N)$, and S_n denote the cyclic shift, where $n \in \{1, 2, \ldots, N\}$, N is the number of orientation channels. When the input X rotates, the output can be expressed as

$$\Phi(P_r^n(X)) = S_n(P_r^n(\Phi(X))), \quad (1 \le n \le N). \tag{3}$$

After a rotation transformation T_r^n on the input X, the rotation-equivariant feature map is transformed by T_r^n that it has a cyclic shift S_n in corresponding channel groups.

C. Analysis

We introduce the circulant matrix to represent the cyclic shift phenomena, i.e., the S_i operation in (5). The circulant matrix is denoted as $\mathcal{C}^n \in \mathbb{R}^{N \times N}$, as

$$\mathcal{C}^n = \begin{bmatrix} 0 & I_{N-n} \\ I_n & 0 \end{bmatrix}, \quad (1 \le n \le N) \tag{4}$$

where I_n denotes a n-dimensional unit matrix.

Supposing there are two feature vectors within channel groups, $V_1 = [v_1, v_2, \ldots, v_N]$ can be cyclically shifted to $V_2 = [v_N, v_1, \ldots, v_{N-1}]$ via $\mathcal{C}^1$ as

$$V_1 \mathcal{C}^1 = V_2. \tag{5}$$

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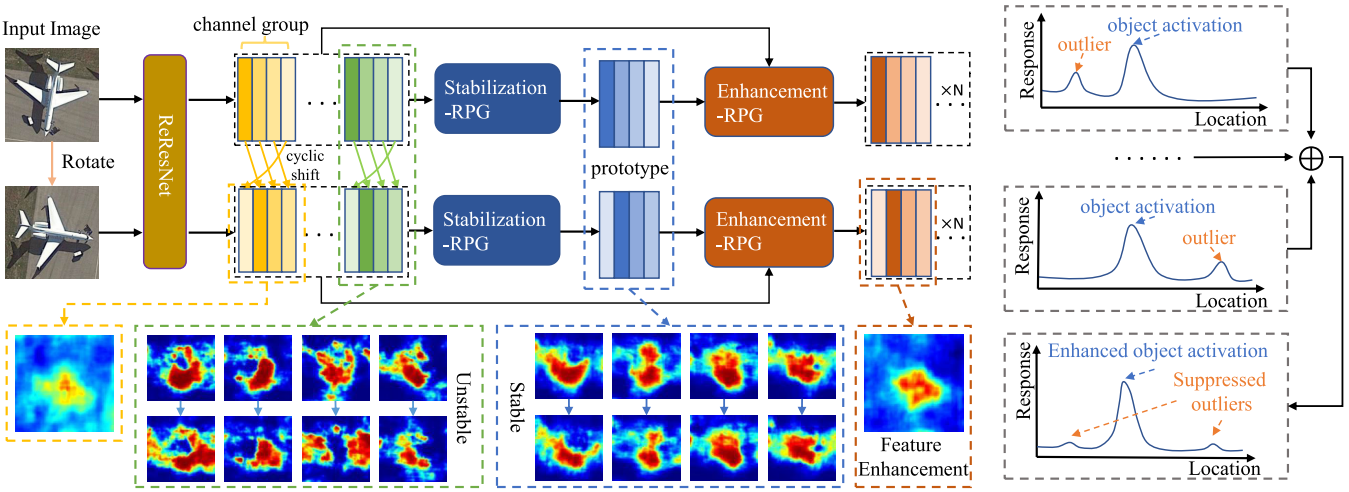


Fig. 2. Pipeline of our method. The feature maps after the rotation-equivariant network have cyclic shift phenomena in each channel group, but they are unstable. We design the stabilization-RPG module to generate the prototype feature which has a stable cyclic shift. In the enhancement-RPG module, the prototype feature is used to enhance the orientation semantics for each channel group. The right side illustrates an explanation of the rotation-robust prototype, wherein the stabilization module suppresses random outliers within each channel group, and location denotes positions on each channel. The response of the target position is generally higher in each channel, while noise occurs randomly at each position. Therefore, our RPG model could serve to smooth out the noise.

In (5), S_n and T_r^n $(1 \leq n \leq N)$ are in one-to-one correspondence since T_r^n is directly related to the actual orientation of the object, so it is important that the feature of an object could maintain the correlation $\mathcal{C}^n(1 \leq n \leq N)$ before and after rotation. Theoretically, when an object rotates, the channels of its corresponding rotation-equivariant features are strictly cyclically shifted. However, due to the dilation and pooling operations inherent in most backbones, this causes artifacts at the feature, making it difficult to maintain rotation-equivariant [37]. Consequently, it leads to the instability of the cyclic shift C^n among channels, as depicted in Fig. 2.

So in the rotation-equivariant CNNs, we observe

$$F\mathcal{C}^1 \neq F_{\text{rot}} \tag{6}$$

where F denotes the features generated by the network before object rotation, which contains only one channel group, $F = [f_1, f_2, \ldots, f_N]$. F_{rot} denotes the features generated by the network after object rotation, $F_{\text{rot}} = [f_N^r, f_1^r, \ldots, f_{N-1}^r]$.

To solve this problem, we design a rotation-equivariance prototype model to activate the orientation feature among channels, satisfying the cyclic shift in rotation-equivariant CNNs.

IV. Method

The RPG method consists of two main parts, namely, the stabilization module and the enhancement module. In the stabilization module, we generate the Rotation-robust prototype. In the enhancement module, the channel group learns residuals from the prototype to augment the semantics of the features.

Section IV-A presents an introduction to stable modules in the RPG approach. The description of the enhancement module in the RPG method is provided in Section IV-B. Section IV-C then discusses the application of this method in both single-stage and two-stage networks.

A. Channel Shift Stabilization

The generation process of the rotation-robust prototype could be concisely formulated as follows:

$$x_{\text{pro}} = \phi(\psi(x, K)) \tag{7}$$

where $\psi(x; K)$ denotes the slice of the input x into K parts, $\phi(\cdot)$ denotes the aggregation operation of the results and x_{pro} denotes the rotation-robust prototype.

We evenly split the feature $x \in \mathbf{R}^{\mathbf{H}\times\mathbf{W}\times\mathbf{C}}$ into K sliced features based on the number of channels, where H, W, and C represent the height, width, and channels of the feature, respectively. Consequently, the dimensions of these K sliced features after partitioning are all $[H, W, (C/K)]$, where C is divisible by K.

Theoretically, there are many ways to aggregate features. We use summation as the most fundamental way and find it simple but really effective. Now $\phi(\cdot)$ can be expressed as

$$\phi(\cdot) = \sum_{k=0}^{K} x_k \tag{8}$$

where x_k represents the kth element generated by $\psi(x; K)$.

As shown in the right part of Fig. 2, we consider channel group features in one-dimension. Each channel group feature has a high response to the object location of interest and also contains random outliers. The involvement of the rotation-robust prototype suppresses the misactivated region (i.e., those that do not belong to the original object but are misactivated when orientation changes), thus making the model more stable to rotation.

B. Semantic Enhancement

The prototype contains the stable circular shift, which preserves the significant semantic information in each channel group. We employ the prototype feature to enhance rotation-equivariant features in each channel group.

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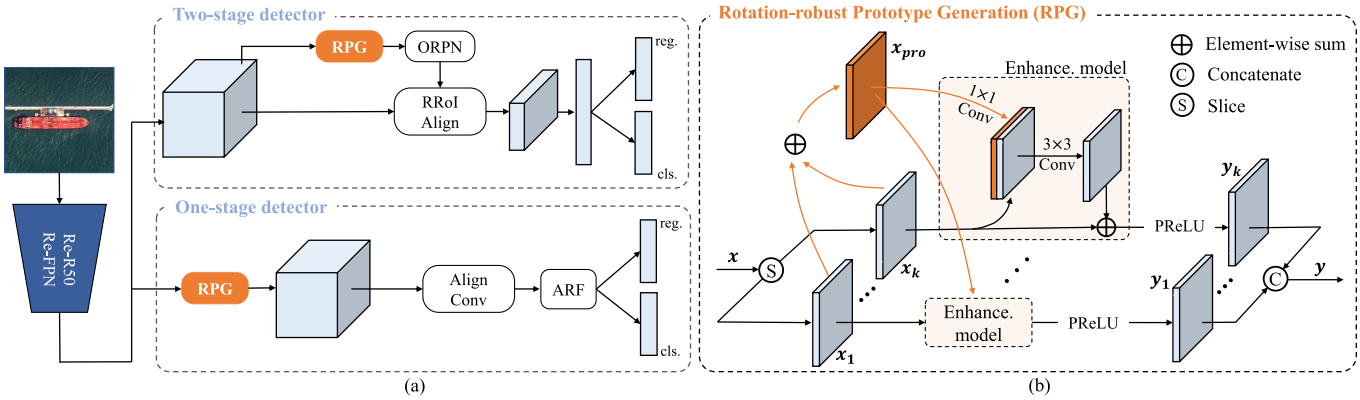


Fig. 3. Pipeline of networks and the details of the RPG model. (a) Proposed RPG module can be plugged into both one-stage and two-stage networks. The RPG model is plugged into the RPN in two-stage detectors. The RPG model is plugged between the rotation-equivariant backbone and the detection head in one-stage detectors. (b) Complete RPG model.

Initially, we apply a convolutional layer to the prototype to narrow the numerical gap between it and the sliced features. We then extract complementary semantic features using convolution by separately concatenating prototypes with each sliced feature. Finally, we add the obtained semantic features to the sliced features to achieve the enhancement purpose. The process can be expressed as

$$y_k = \text{prelu}(\text{conv}(\text{cat}(x_k, \text{conv}(x_{\text{pro}}))) + x_k) \tag{9}$$

where $\text{cat}(\cdot,\cdot)$ denotes that features are concatenated in channel dimension, $\text{conv}(\cdot)$ denotes a conventional convolutional layer, $+$ denotes element-wise summation, and $\text{prelu}(\cdot)$ denotes the nonlinear activation function.

We concatenate all the channel group features y_k in the channel dimension to get the output feature y

$$y = \text{cat}(y_1, y_2, \ldots, y_k). \tag{10}$$

By learning stable cyclic shifts from the prototype, each x_k is able to acquire better orientation semantics. Also, since the prototype has the "standard" semantics of all group channels, x_k could learn the correlation among group channels to activate the semantics implicitly in itself.

C. Plug Into Detectors

The fact that the dimensions of input x and output y are the same, which ensures that the RPG can be plug-and-play in different frameworks of detectors, as shown in Fig. 3. We incorporated our module into both one-stage and two-stage detectors.

First, we use S^2ANet [16] as the baseline of the one-stage detector. The original ResNet50-FPN [19], [28] backbone is replaced with the equivariant backbone, i.e., ReResNet50-ReFPN [17], to extract rotation-equivariant features. Then, we embed the RPG module following ReFPN. For S^2ANet, anchor boxes are generated by ARN, and the features are aligned by the alignment convolution according to the anchor boxes. After that, the ARF is used to encode orientation information for detection.

In the two-stage detector, the oriented R-CNN [49] is used as the baseline. Similar to the aforementioned one-stage detector, the original ResNet50-FPN [19], [28] backbone is replaced by the ReResNet50-ReFPN [17] to extract rotation-equivariant features. Our RPG module is embedded into the RPN stage and RoI is generated by using midpoint offset representation. Then, it is fine-tuned in the second stage for localization and classification.

Since both networks are anchor-based, we incorporate the RPG module into the one-stage anchor-free network, FCOS [42]. In a manner similar to the S^2ANet approach, we replace the original backbone of the network and subsequently embed the RPG module after the ReFPN.

V. Experiments and Analysis

A. Datasets

We conduct experiments on four widely used datasets: DOTA-v1.5 [10], DOTA-v1.0 [48], HRSC2016 [32], and DIOR-R [5].

1) **DOTA-v1.0** is a large dataset for oriented object detection in aerial images. It contains 2806 large remote sensing images and 188 282 instances with OBB annotations among 15 categories: tennis court (TC), small vehicle (SV), soccer-ball field (SBF), harbor (HA), ship (SH), ground track field (GTF), plane (PL), helicopter (HC), storage tank (ST), swimming pool (SP), baseball diamond (BD), large vehicle (LV), roundabout (RA), basketball court (BC), and bridge (BR). The size of the image ranges from 800 × 800 to 4000 × 4000.
2) **DOTA-v1.5** is a large-scale dataset for object detection in aerial images. It is an expansion based on the DOTA-v1.0 dataset, introducing the category of container crane. It comprises 2806 aerial images, encompassing 16 categories with a total of 400 000 instances. We use training and validation sets for training and the test set for testing.
3) **HRSC** is another widely used ship detection dataset. A total of 1061 remote sensing images are included, of which 436 are for the training set, 181 for the validation set, and 444 for the test set. Image sizes range from 300 × 300 to 1500 × 900. Similarly, we use the training set the validation set for training, and the test set for testing.

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4) **DIOR-R** is based on DIOR [26], which includes 23 463 images and 192 518 instances, covering a wide range of scenes and 20 common object classes. The categories of objects in DIOR-R include windmill (WM), vehicle (VE), train station (TS), tennis court (TC), storage tank (STO), stadium (STA), ship (SH), overpass (OP), harbor (HA), ground track field (GTF), golf field (GF), dam (DAM), expressway toll station (ETS), expressway service area (ESA), chimney (CH), bridge (BR), basketball court (BC), baseball field (BF), airport (APO), and airplane (APL).

B. Parameter Settings

We use 2 NVIDIA RTX 2080Ti GPUs with a total batch size of 8 for training and a single RTX 2080Ti GPU for inference. All our experiments are based on the MMDetection benchmark [3]. ReResNet50 [17] and ReFPN [17] are used as our equivariant feature extractors. ReResNet is pretrained on ImageNet. Data augmentation during training uses horizontal and vertical flips. We optimize the overall network with the SGD algorithm with a momentum of 0.9 and a weight decay of 0.0001. For the DOTA-v1.0 dataset, we crop the original images into 1024 × 1024 patches. Similar to other methods [5], [17], the clipping stride of the images is set to 824, that is to say, the pixel overlap between two adjacent patches is 200. Unlike other methods, we trained equivariant-oriented R-CNN with 24 epochs. This is illustrated later in the experiment. The initial learning rate is set to 0.005 and divided by 10 at epochs 16 and 22. The ploy NMS threshold is set to 0.1 when merging image patches. For the DIOR-R dataset, we keep the image size at the original size of 800 × 800. The same training schedule as the DOTA dataset is implemented. For the HRSC2016 dataset, we resize the image to 800 × 800. Similar to the configuration widely used by other methods, we trained the network with 36 epochs. The initialization learning rate is set to 0.005 and divided by 10 at epochs 24 and 33.

C. Ablation Study

In this section, we conduct ablation studies on the parameters of the RPG module. Unless otherwise specified, ablation experiments are performed on DOTAv-1.0. Specifically, we train the network on the training and validation sets and conduct testing on the test set.

At first, we analyze the effect of the number of channel slices in the stabilization module, as shown in Table I. When K is set to 16 and 32, we can achieve 76.23% and 76.28% mAP, respectively. Neither of the above is a good improvement. When setting K to 8, there is a large performance improvement of 0.89%. The summation operation has a tradeoff between preserving semantic information and reducing feature noise. If the channels are split into too few sliced features (i.e., K is small), the summation cannot effectively filter out outliers and noise, leading to performance degradation. If the channels are split into too many sliced features (i.e., K is very large), the summation may also diminish the semantic information. Therefore, we choose K to be 8 as a reasonable compromise.

TABLE I

EXPERIMENTS ABOUT CHANNEL SLICE (K). NOTE THAT A LARGER K MEANS FEWER CHANNELS PER SLICED FEATURE AND MORE SEMANTIC LOSS

Method	Channel slice (K)	mAP
RPG-Ori R-CNN(ours)	32	76.28
	16	76.23
	8	**77.17(+0.89)**
	4	75.49

TABLE II

ABLATIONS OF CONVOLUTION KERNEL SIZE. THE NETWORK USES SMOOTHL1LOSS AS THE LOCATION LOSS

Method	kernel size	mAP
RPG-Reina-O(ours)	1×1	71.99
	3×3	**72.27(+0.28)**
	5×5	71.60

TABLE III

RESULTS OF USING RPG ON TWO-STAGE DETECTOR. ORI R-CNN INDICATES THE ORIENTED R-CNN AND E-ORI R-CNN INDICATES THE ORIENTED R-CNN WITH AN EQUIVARIANT BACKBONE. RPG-ORI R-CNN INDICATES ORIENTED R-CNN WITH OUR RPG

Method	Epoch	Backbone	RPG	mAP
Ori R-CNN	12	R-50		75.87
Ori R-CNN	24	R-50		75.59
E-Ori R-CNN	12	ReR-50		74.97
E-Ori R-CNN	24	ReR-50		76.04(+0.17)
RPG-Ori R-CNN (ours)	24	ReR-50	✔	**77.17(+1.13)**

TABLE IV

RESULTS OF USING RPG ON ONE-STAGE DETECTOR. E-S^{2}ANET INDICATES S^{2}ANET WITH EQUIVARIANT BACKBONE AND RPG-S^{2}ANET INDICATES S^{2}ANET WITH OUR RPG

Method	Epoch	Backbone	RPG	mAP
S^{2}ANet	12	R-50		74.12
S^{2}ANet	24	R-50		75.44
RPG-S^{2}ANet (ours)	24	R-50	✔	76.00(+0.56)
E-S^{2}ANet	24	ReR-50		75.51(+0.07)
RPG-S^{2}ANet (ours)	24	ReR-50	✔	**76.49(+0.98)**

Furthermore, we analyze the size of the convolution kernel used in the enhancement module feature extraction part, as shown in Table II. When 1×1 convolution is used, the performance of the network achieves 71.99% mAP. However, since rotations usually occur at the object-level, neighboring depth pixels have the same circular shift correlation. After replacing the 1×1 convolution with a 3×3 convolution, the performance gains by 0.28%. It drops after using 5×5 convolution. This is due to the fact that the rotation-equivariant property in aerial images is at the instance-level. This implies that neighboring depth pixels on the feature map tend to share the same rotation, so the convolution kernel of size 3 can better capture the rotation features.

D. Effectiveness of RPG

In this section, we incorporate the RPG module into different networks to evaluate its effectiveness and report its computational cost. For the rotation-equivariant backbone, we use the same setting as in ReDet [17].

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TABLE V
RESULTS OF USING RPG ON DIOR-R. ORI R-CNN INDICATES THE ORIENTED R-CNN

Method	Epoch	Backbone	RPG	mAP
S^2ANet	12	R-50		62.7
E-S^2ANet	24	R-50		65.0
RPG-S^2ANet (ours)	24	R-50	✔	67.8(+2.8)
Ori R-CNN	12	R-50		65.7
E-Ori R-CNN	24	ReR-50		65.2
RPG-Ori R-CNN (ours)	24	ReR-50	✔	66.6(+1.4)

TABLE VI
RESULTS OF USING RPG ON DOTA-V1.5. ORI R-CNN INDICATES THE ORIENTED R-CNN

Method	Epoch	Backbone	RPG	mAP
S^2ANet	12	R-50		61.7
E-S^2ANet	24	R-50		64.3
RPG-S^2ANet (ours)	24	R-50	✔	66.2(+1.9)
Ori R-CNN	12	R-50		65.9
E-Ori R-CNN	24	ReR-50		66.3
RPG-Ori R-CNN (ours)	24	ReR-50	✔	66.7(+0.4)

TABLE VII
RESULTS OF USING RPG ON ANCHOR-FREE DETECTOR. E-FCOS-OBB INDICATES FCOS-OBB WITH EQUIVARIANT BACKBONE AND RPG-FCOS-OBB INDICATES RPG-FCOS-OBB WITH OUR RPG

Method	Epoch	Backbone	mAP
FCOS-OBB	12	R-50	69.81
FCOS-OBB	24	R-50	73.49
E-FCOS-OBB	24	ReR-50	73.31
RPG-FCOS-OBB (our)	24	ReR-50	**74.25(+0.94)**

TABLE VIII
JOINT USE OF CHANNEL ALIGNMENT AND RPG. †INDICATES THAT THE RESULTS ARE REIMPLEMENTED

Method	C-Align.	RPG	mAP
ReDet†	✔		75.71
ReDet†	✔	✔	76.28(+0.57)

TABLE IX
COMPLEXITY OF THE MODEL

Method	Flops (G)	Params (M)	FPS
E-S^2ANet	67.77	8.62	0.59
RPG-S^2ANet(ours)	71.02(+3.25)	8.77(+0.15)	0.59

We first evaluate our module on the DOTA-v1.0. In the two-stage network, the results are shown in Table III. The original Ori. R-CNN reaches its peak performance when trained for 12 epochs, with a mAP of 75.87%, and degrades to 75.59% when trained for 24 epochs. When we replace its backbone with a rotation-equivariant network, the opposite trend is observed, and the best performance is achieved when trained for 24 epochs, with a mAP of 96.04%. We can see that changing the backbone only brings marginal improvement to the network. By inserting the RPG module, the network is able to extract stable rotation representation and enhance features, resulting in a considerable boost in performance, with a mAP of 77.17%.

In the one-stage network, we first test the impact of training batches on the network. The results are shown in Table IV. Unlike the two-stage network, the baseline network (S^2ANet) achieves its best performance when trained for 24 epochs, with an mAP of 75.44%. When we replace its backbone with a rotation-equivariant network, the detector shows a slight improvement, with a mAP of 75.51%. When we add the RPG module, the detector exhibits a significant improvement as in the two-stage case and reaches an mAP of 76.49%. This demonstrates that the RPG module can be effective in both one-stage and two-stage networks. Interestingly, we also apply the RPG module to the original S^2ANet and observe some improvement over the baseline, with an mAP of 76.00%. The reason may be that the original CNN has limited encoding capacity for orientation, which prevents the network from obtaining stable rotation representations. The improvement in mAP may be attributed to the summation operation suppressing outliers.

We conduct experiments with RPG-S^2ANet and Ori. R-CNN on the DIOR-R dataset. The results shown in Table V prove the effectiveness of our approach. S^2ANet achieves an mAP of 62.7% without employing equivariant backbones. There is a slight improvement to 65.0% with the use of equivariant backbones. Upon integrating the RPG module, there is a remarkable performance improvement, elevating the mAP to 67.8%. Similarly, RPG demonstrates favorable performance for Ori. R-CNN. Without using equivariant backbones, the network achieves a performance of 65.7% mAP. Surprisingly, incorporating equivariant backbones leads to a slight decrease in network performance. However, with the addition of the RPG module, the network shows a positive enhancement, reaching 66.6% mAP.

To further substantiate the effectiveness of our RPG, we conduct experiments on DOTA1.5. We train on the training set and test on the validation set. As shown in Table VI, without using rotation-equivariant backbones, S^2ANet and Ori. R-CNN achieved performances of 61.7% and 65.9%, respectively. With the utilization of the equivariant backbone, both models exhibit a slight improvement in mAP, reaching 64.3% and 66.3%. However, with the incorporation of the RPG module, there is a significant enhancement for the one-stage network, pushing the mAP to 66.2%.

Since Ori. RCNN and S^2ANet are both anchor-based methods, to further verify the effectiveness of the RPG module, we use the FCOS-OBB network (an anchor-free method) as the baseline to test the RPG module. The results are shown in Table VII. The baseline network reaches the peak performance when trained for 24 epochs, with a mAP of 73.49%. When we replace the backbone with a rotation-equivariant network, the detector's performance deteriorates to 73.31%. We speculate that this is because it introduces excessive noise, which impairs the detection quality. When we add the RPG module, the network achieves a noticeable improvement and attains a mAP of 74.25%. This indicates that the RPG module is still effective for anchor-free detectors.

In the two-stage network, some methods [17] employ a rotation-equivariant network to extract rotation-equivariant

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TABLE X

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DOTA-V1.0 DATASET. H-104 MEANS HOURGLASS 104 [24]

Method	Backbone	PL	BD	BR	GTF	SV	LV	SH	TC	BC	ST	SBF	RA	HA	SP	HC	mAP
One-stage																	
PIoU [4]	DLA-34	80.90	69.70	24.10	60.20	38.30	64.40	64.80	**90.90**	77.20	70.40	46.50	37.10	57.10	61.90	64.00	60.50
RetinaNet-O	R50-FPN	88.67	77.62	41.81	58.17	74.58	71.64	79.11	90.29	82.18	74.32	54.75	60.60	62.57	69.67	60.64	68.43
DRN [38]	H-104	88.91	80.22	43.52	63.35	73.48	70.69	84.94	90.14	83.85	84.11	50.12	58.41	67.62	68.60	52.50	70.70
DAL [36]	R50-FPN	88.68	76.55	45.08	66.80	67.00	76.76	79.74	90.84	79.54	78.45	57.71	62.27	69.05	**73.14**	60.11	71.44
RSDet [39]	R101-FPN	**89.80**	82.90	48.60	65.20	69.50	70.10	70.20	90.50	85.60	83.40	62.50	63.90	65.60	67.20	**68.00**	72.20
DCFL [50]	R50-FPN	89.22	79.15	50.95	71.29	77.44	73.92	84.22	90.90	84.07	84.73	57.07	66.53	65.84	69.51	55.09	73.33
R^3Det [53]	R101-FPN	88.76	83.09	50.91	67.27	76.23	80.39	86.72	90.78	84.68	83.24	61.98	61.35	66.91	70.63	53.94	73.79
S^2ANet [16]	R50-FPN	89.11	82.84	48.37	71.11	78.11	78.39	87.25	90.83	84.90	85.64	60.36	62.60	65.26	69.13	57.94	74.12
Ori. RepPoints [27]	R50-FPN	87.02	**83.17**	54.13	71.16	**80.18**	78.40	**87.28**	**90.90**	85.97	**86.25**	59.90	**70.49**	73.53	72.27	58.97	75.97
RPG-S^2ANet(ours)	ReR50-ReFPN	89.29	81.96	**54.29**	**74.05**	77.59	**82.43**	87.26	90.72	**87.25**	85.75	**64.02**	67.11	**75.54**	69.91	60.14	**76.49**
Two-stage																	
ICN [1]	R101-FPN	81.36	74.30	47.70	70.32	64.89	67.82	69.98	90.76	79.06	78.20	53.64	62.90	67.02	64.17	50.23	68.16
Faster R-CNN-O	R50-FPN	88.44	73.06	44.86	59.09	73.25	71.49	77.11	90.84	78.94	83.90	48.59	62.95	62.18	64.91	56.18	69.05
RoI Transformer [9]	R101-FPN	88.64	78.52	43.44	75.92	68.81	73.68	83.59	90.74	77.27	81.46	58.39	53.54	62.83	58.93	47.67	69.56
CAD-Net [58]	R101-FPN	87.80	82.40	49.40	73.50	71.10	63.50	76.60	**90.90**	79.20	73.30	48.40	60.90	62.00	67.00	62.20	69.90
CenterMap-Net [44]	R50-FPN	88.88	81.24	53.15	60.65	78.62	66.55	78.10	88.83	77.80	83.61	49.36	66.19	72.10	**72.36**	58.70	71.74
SCRDet [55]	R101-FPN	89.98	80.65	52.09	68.36	68.36	60.32	72.41	90.85	**87.94**	**86.86**	65.02	66.68	66.25	68.24	65.21	72.61
FAOD [25]	R101-FPN	**90.21**	79.58	45.49	76.41	73.18	68.27	79.56	90.83	83.40	84.68	53.40	65.42	74.17	69.69	64.86	73.28
FR-Est [13]	R101-FPN	89.63	81.17	50.44	70.19	73.52	77.98	86.44	90.82	84.13	83.56	60.64	66.59	70.59	66.72	60.55	74.20
RoI Transformer$^+$	R50-FPN	88.65	82.60	52.53	70.87	77.93	76.67	86.87	90.71	83.83	82.51	53.95	67.61	74.67	68.75	61.03	74.61
Mask OBB [43]	R50-FPN	89.61	**85.09**	51.85	72.90	75.28	73.23	85.57	90.37	82.08	85.05	55.73	68.39	71.61	69.87	**66.33**	74.86
Gliding Vertex [52]	R101-FPN	89.64	85.00	52.26	**77.34**	73.01	73.14	86.82	90.74	79.02	86.81	59.55	70.91	72.94	70.86	57.32	75.02
Oriented R-CNN [49]	R50-FPN	89.46	82.12	54.78	70.86	**78.93**	83.00	**88.20**	**90.90**	87.50	84.68	63.97	67.69	74.94	68.84	52.28	75.87
ReDet [17]	ReR50-ReFPN	88.79	82.64	53.97	74.00	78.13	**84.06**	88.04	90.89	87.78	85.75	61.76	60.39	**75.96**	68.07	63.59	76.25
Oriented R-CNN [49]	R101-FPN	88.86	83.48	**55.27**	76.92	74.27	82.10	87.52	**90.90**	85.56	85.33	65.51	66.82	74.36	70.15	57.28	76.28
RPG-Ori R(ours)	ReR50-ReFPN	89.90	83.92	53.72	70.99	78.20	83.42	88.04	90.88	86.72	85.94	**67.21**	**68.47**	75.66	72.10	62.38	**77.17**

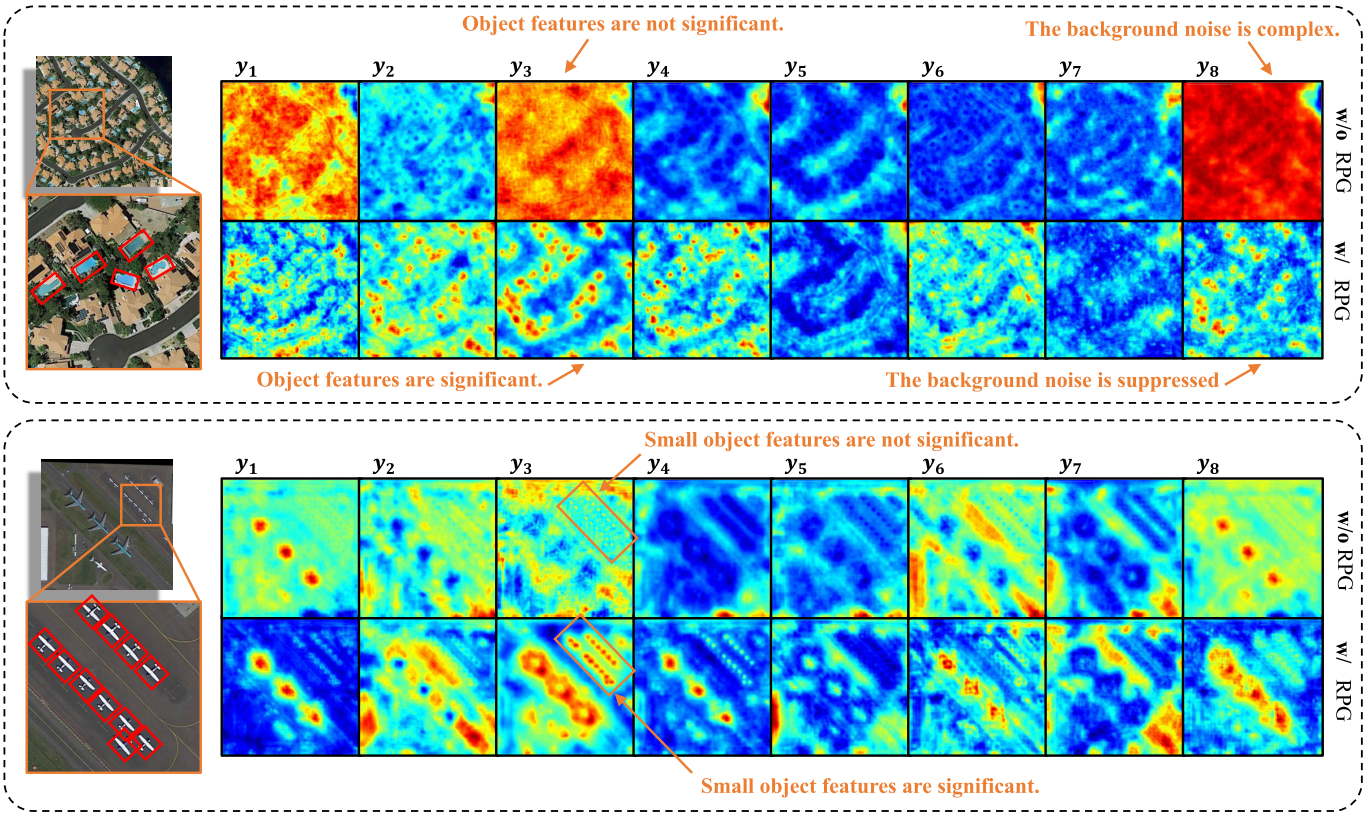


Fig. 4. Detection of small objects. y_i denotes the ith channel group of the feature ($1 \leq i \leq 8$). The feature maps in the first row are generated by the network without RPG. The maps in the second row are generated by the network with RPG.

features and use channel alignment [17] to acquire rotation-invariant features in the second stage. In Table VIII, we evaluate the combination of the RPG module and the channel alignment module, and the results demonstrate that the RPG module remains effective when the network adopts channel alignment. This is because applying the RPG module in the RPN stage does not interfere with the input features of the ROI Head stage, allowing the channel alignment module to still extract rotation-invariant features.

Since the RPG module is a plug-and-play component, we measure its computational cost, and the results are presented in Table IX. By adding the RPG module, Flops and Params increase by 3.25G and 0.15M, respectively. Since the

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TABLE XI
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DOTA-V1.5 DATASET

Method	S^2ANet [16]	DCFL [50]	Ori R-CNN [49]	RPG-S^2ANet (ours)	RPG-Ori R-CNN (ours)
mAP	61.7	63.5	65.9	66.2	**66.7**

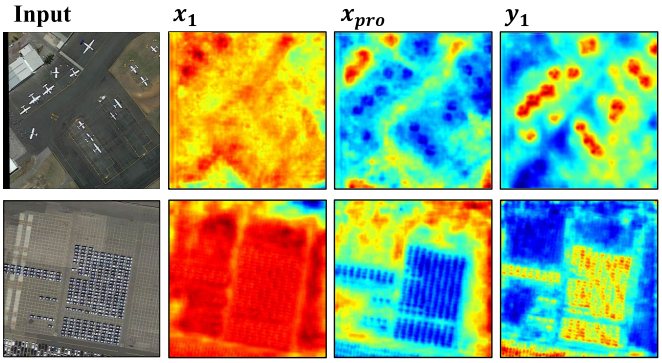


Fig. 5. Visualization of feature maps in RPG. The individual channel groups (e.g., x_1) of rotation-equivariant features have a lot of noise. RPG slices and recombines the channels to generate prototype features (e.g., x_{pro}) that reflect the key semantics of the features. The channels can also reduce the impact of noises by learning with the prototype features.

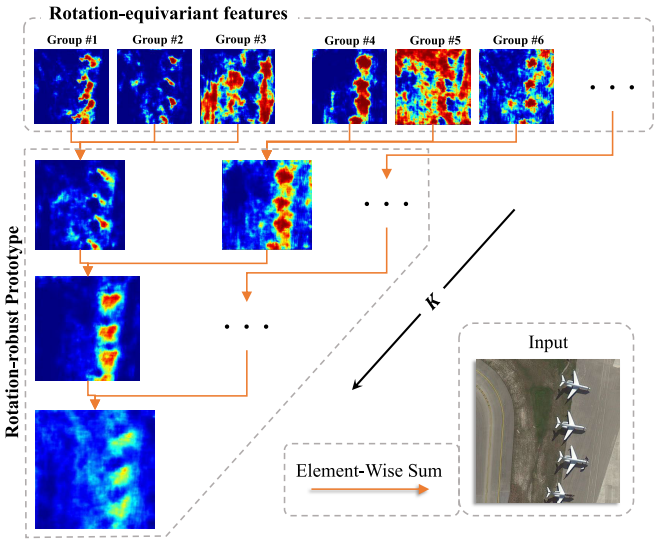


Fig. 6. Illustration of the generation of the prototype. In the stabilization module, we use the parameter K to represent the degree of feature slicing and use summation to aggregate features. While summation can suppress outliers, it can also result in semantics loss of objects. When K is larger, the degree of feature slicing is larger and the number of summations is higher, which is not useful for semantics preservation. Therefore, the features need to be appropriately sliced.

rotation-equivariant backbone itself has lower computational overhead than the original CNN, the increase in computation caused by the RPG module is tolerable.

E. Analysis

In this section, we conduct a visual analysis and discussion of the network features.

Aerial images present a significant challenge due to their abundant small objects that are often difficult to detect accurately. Our proposed method addresses this issue by leveraging the rotation-robust prototype to learn stable orientation semantics and effectively suppress background outliers. As a result, the localization of small objects is notably improved, benefiting from more stable channel shifts and enhanced object semantics. For instance, in the top example of Fig. 4, our RPG method accurately localizes swimming pools. By examining the activation of channel groups, it becomes evident that the swimming pool features are highlighted while the background noise is effectively suppressed.

Furthermore, the bottom example of Fig. 4 demonstrates that, when the RPG module is not used, the network primarily emphasizes large aircraft, while our RPG module successfully reactivates and captures those regions corresponding to small objects.

These results demonstrate that our RPG module effectively enhances the performance of the detector, particularly in the challenging context of aerial images with numerous small objects. By leveraging rotation-robust representations and suppressing background noise, our method achieves more robust and accurate object detection, making it well-suited for aerial imagery applications.

We further visualize the prototype features, and the results are shown in Fig. 5. In the original rotation-equivariant features, some channels exhibit abnormal responses, where foreground and background features become confounded. However, the prototype features reveal distinct rotation responses between foreground and background, and this contrast between them strengthens as the training progresses. This is attributed to the inherent orientation differences between foreground and background.

In the enhancement model, by learning from the prototype features, the sliced features are capable of suppressing regions with abnormal responses in each channel, while emphasizing the responses of target objects. Consequently, this enhances the overall detection performance.

Discussion: We slice and aggregate the channel groups to obtain the prototype. The experiments in Table I performed an ablation study on K. Furthermore, we visualize the effect of different summation operations in Fig. 6, and the results show that summation among moderate groups is beneficial for enhancing the object response. Theoretically, the use of RPG has not been limited to the rotation-equivariant networks. For conventional CNNs, the prototype features generated by the stabilization module contain the semantic information in all channel groups and are similarly suppressive for outliers. In the stabilization module, each channel group can obtain the semantic information of other channel groups from the prototype to complement its own semantic information. RPG differs from the attention mechanism in that the attention mechanism performs the enhancement in a bottom-up manner, applying weight masks to each position in the feature map. However, RPG enhances the feature semantics with the help of the generated prototypes, which contain more interpretable semantic information and could help refine the original features in a top-down manner.

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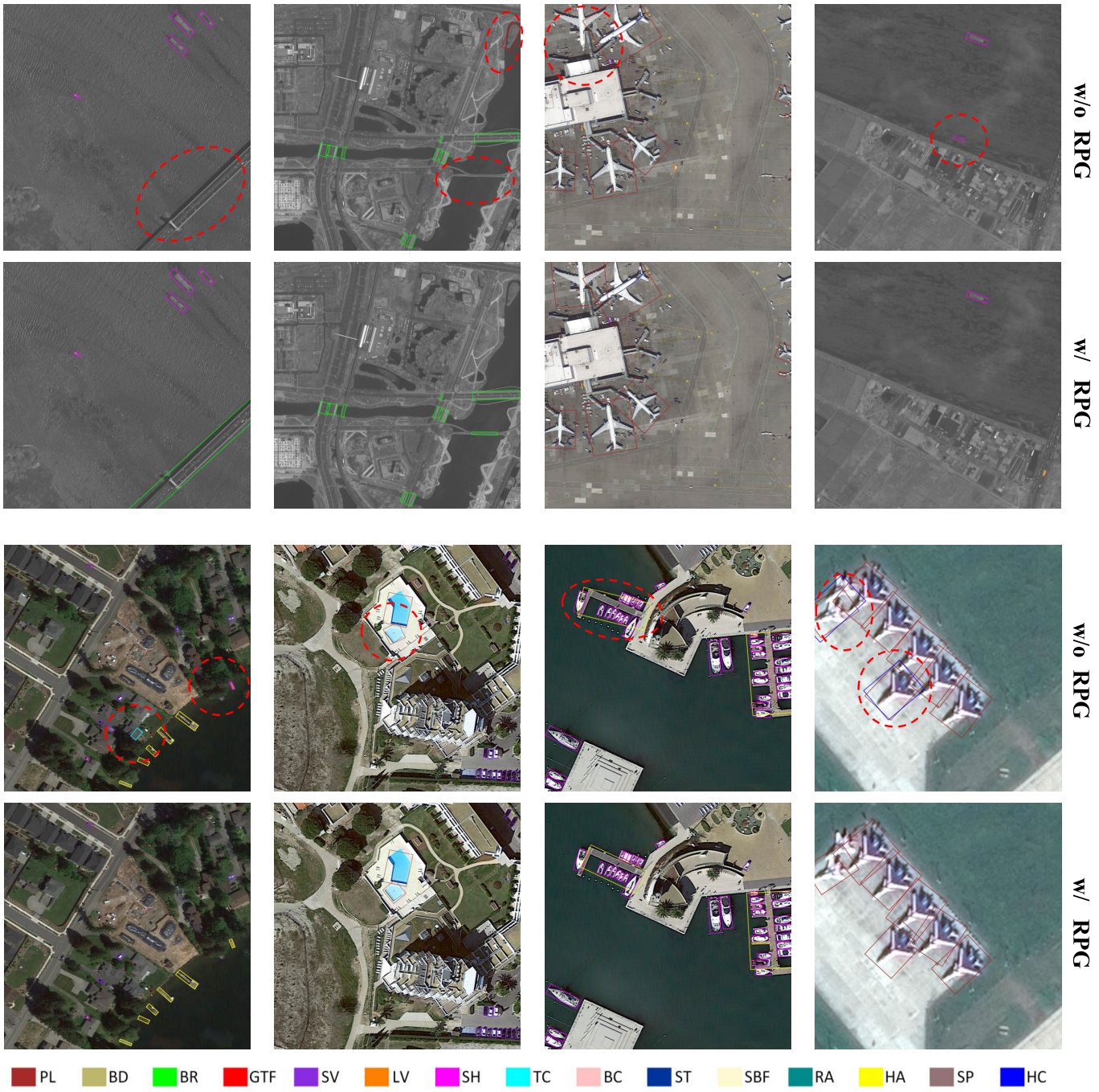


Fig. 7. Detection results on DOTA-v1.0. The confidence threshold is set to 0.3 for visualization. One color stands for one category. The red dashed lines encircle the targets with false positives, missed detections, and incorrect orientation recognition.

TABLE XII

COMPARISON WITH STATE-OF-THE-ART METHODS ON THE DIOR-R DATASET

Method	APL	APO	BF	BC	BR	CH	DAM	ETS	ESA	GF	GTF	HA	OP	SH	STA	STO	TC	TS	VE	WM	mAP
Faster RCNN-O	62.79	26.8	71.72	80.91	34.2	72.57	18.95	66.45	65.75	66.63	79.24	34.95	48.79	81.14	64.34	71.21	81.44	47.31	50.46	65.21	59.54
RetinaNet-O	61.49	28.52	73.57	81.17	23.98	72.54	19.94	72.39	58.2	69.25	79.54	32.14	44.87	77.71	67.57	61.09	81.46	47.33	38.01	60.24	57.55
Gliding Vertex [52]	65.35	28.87	74.96	81.33	33.88	74.31	19.58	70.72	64.7	72.3	78.68	37.22	49.64	80.22	69.26	61.13	81.49	44.76	47.71	65.04	60.06
RoI Trans [9]	63.34	37.88	71.78	87.53	40.68	72.6	26.86	**78.71**	68.09	68.96	82.74	**47.71**	55.61	81.21	**78.23**	70.26	81.61	54.86	43.27	65.52	63.87
AOPG [5]	62.39	37.79	71.62	87.63	40.90	72.47	31.08	65.42	77.99	73.20	81.94	42.32	54.45	81.17	72.69	**71.31**	81.49	**60.04**	52.38	**69.99**	64.41
Ours																					
RPG-Ori R-CNN	63.40	46.39	71.81	**89.47**	45.34	**79.64**	30.20	70.75	80.35	**75.65**	**82.84**	47.57	58.64	81.10	66.56	62.57	**89.58**	56.45	43.53	66.05	65.39
RPG-S^{2}ANet	**71.35**	**51.66**	**76.57**	87.84	**46.82**	73.20	**31.45**	74.23	**84.07**	72.02	82.20	46.57	**59.65**	**85.90**	60.71	70.01	88.62	59.30	**53.51**	65.32	**67.05**

F. Comparison With State-of-the-Arts

In this section, we compare our method with other competitors on four datasets: DOTA-v1.0, DOTA-v1.5, DIOR-R, and HRSC2016.

First, we evaluate the performance of RPG-S^{2}ANet and RPG-Ori. R-CNN on the DOTA-v1.0 dataset. This dataset presents a challenging detection benchmark with numerous object categories and diverse scenes. As shown in Table X,

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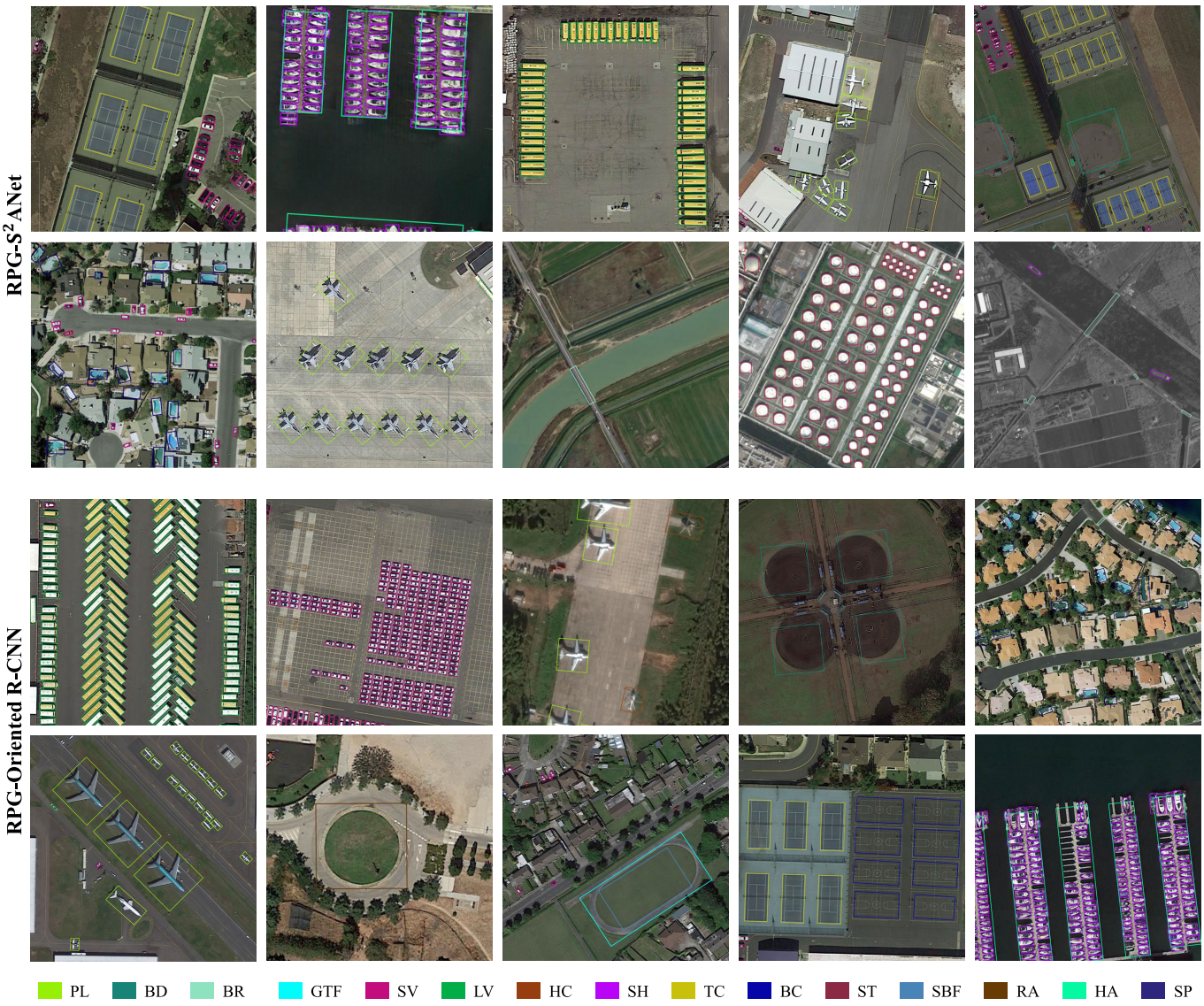


Fig. 8. Detection results on DOTA-v1.0. The confidence threshold is set to 0.3 for visualization. One color stands for one category.

TABLE XIII
COMPARISON WITH STATE-OF-THE-ART METHODS ON THE HRSC2016 DATASET

Method	Backbone	mAP
PIoU [4] (*2020, ECCV*)	DLA-34	89.20
R^3Det [53] (*2021, AAAI*)	R101-FPN	89.26
DAL [36] (*2021, AAAI*)	R101-FPN	89.77
S^2ANet [16] (*2021, TGRS*)	R101-FPN	90.17
Rot. RPN [34] (*2018, TMM*)	R101	79.08
R2CNN [23] (*2017, arXiv*)	R101	73.07
RoI Trans. [9] (*2019, CVPR*)	R101-FPN	86.20
Glid. Ver. [52] (*2020, TPAMI*)	R101-FPN	88.20
Ori. R-CNN [49] (*2021, ICCV*)	R50-FPN	90.40
Ori. RepPoints [27] (*2022, CVPR*)	R-50-FPN	90.38
RPG-Ori R-CNN (ours)	ReR50-ReFPN	90.43
RPG-S^2ANet (ours)	ReR50-ReFPN	**90.45**

in the one-stage network, our proposed RPG-S^2ANet achieves a remarkable 76.49% mAP, surpassing the relatively newer DCFL and Ori. RepPoints methods. Moreover, we outperform other one-stage detectors in six classes: BR, GTF, LV, BC, SBF, and HA. Two-stage networks generally outperform one-stage networks, where RPG-Ori. R-CNN achieves an impressive 77.17% mAP, outperforming detectors like ReDet using rotation-equivariant backbones and even outperforming Oriented R-CNN using ResNet101 as a backbone. Our method achieves the best results in classes RA and HA and remains competitive in other classes.

On the DOTA-v1.5 dataset, both networks maintain competitive performance. DOTA-v1.5 complements DOTA-v1.0 with annotations from smaller datasets and additional categories. As shown in Table XI, while DCFL surpasses our baseline method S^2ANet, incorporating RPG significantly boosts RPG-S^2ANet's performance to 66.2% mAP, notably higher than DCFL. This enhancement owes to RPG's ability to enrich rotation-equivariant features, effectively utilized by the network.

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```python
class RPG(nn.Module):
    def __init__(self,in_c):
        super(RPG,self).__init__()
        self.base_c = in_c//8
        self.w_conv = nn.Conv2d(self.base_c,self.base_c,kernel_size=1)
        self.in_conv = nn.ModuleList()
        self.out_BNReLU = nn.PReLU()
        for i in range(8):
            conv = nn.Conv2d(self.base_c*2,self.base_c,kernel_size=3,padding=1)
            self.in_conv.append(conv)
    def forward(self,x):
        fuse_feat = self.add_c(x)
        fuse_feat = self.w_conv(fuse_feat)
        feats = []
        for i in range(8): # Residual learning
            feat = x[:,i*self.base_c:(i+1)*self.base_c,:,:]
            feat_ = torch.cat([feat,fuse_feat],dim = 1)
            out_feat = self.in_conv[i](feat_)
            feats.append(out_feat+feat)
        out = self.out_BNReLU(torch.cat(feats,dim = 1))
        return out
    def add_c(self,feat): # Generate category prototype
        out_list = []
        for i in range(8):
            out_ = feat[:,i*self.base_c:(i+1)*self.base_c,:,:]
            out_list.append(out_)
        return sum(out_list)
```

To further demonstrate our method's effectiveness, we conduct comparative experiments on the DIOR-R dataset. DIOR-R extends the DIOR dataset, encompassing 20 common object classes and diverse scene types. As presented in Table XII, our RPG-Ori. R-CNN and RPG-S^{2}ANet achieved significant mAP scores of 65.39% and 67.05%, respectively, surpassing the performance of previous state-of-the-art methods by a substantial margin. Our approach excels in 13 classes, substantially outperforming other competitors.

Finally, we test our networks on HRSC2016. This dataset challenges detection with numerous instances of elongated vessels in arbitrary orientations. As shown in Table XIII, our proposed method achieves competitive mAP scores: RPG-Ori. R-CNN and RPG-S^{2}ANet reach 90.43% and 90.45% mAP, respectively.

As an additional qualitative analysis, we present the comparison results of S^{2}ANet with and without using RPG in Fig. 7. We visualize images from different scenes to demonstrate the impact of RPG on the network's output, which illustrates instances of false positives and false negatives in the absence of RPG due to inherent noise in features like rotation. However, when equipped with RPG, the features become cleaner, reducing the interference of noise on the category, thus lowering the occurrence of false detections and misses. Moreover, the visualizations indicate that with RPG, the network better predicts the orientation of the targets, further validating the superiority of our approach. We present the final visualization performance of our approach in Fig. 8.

VI. Conclusion

In this article, we propose an RPG method with two effective modules: a stabilization module and an enhancement module. In the stabilization module, we generate the rotation-robust prototype while in the enhancement module, we use the obtained prototype to augment the original features, activating the semantic representation of those features that are depressed by changes in orientation. The proposed RPG method could be used as a plug-and-play module in any detector as long as the backbone uses the rotation-equivariant convolution. Experimental results on three challenging datasets demonstrated that the proposed RPG method is really effective, while it is also easy to implement.

Appendix

We implement the RPG module with only 27 lines of codes, our experiments achieve an average 1% improvement on three challenging datasets, including DOTA-v1.5, DOTA-v1.0, DIOR-R, and HRSC2016.

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